Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

School of Computer Science and Engineering

Fall Semester 2025-26

BCSE331P – Exploratory Data Analysis

Assessment 3

Data Cleaning and Univariate Analysis **CO2 BT4**

CO2: Debug and execute R programs using R studio.

Load dplyr, tidyr, ggplot2, stringr, forcats into R

1. Import a dataset into R. (Dataset shared through e-mail)
2. Display the first and last six records of the dataset
3. Identify the total number of missing values in the dataset.
4. Display all records containing at least one missing value.
5. Remove columns having more than 70 percentage of missing values.
6. Replace missing numerical values with the mean.
7. Replace missing numerical values with the median.
8. Replace missing categorical values with the mode.
9. Identify duplicate rows if any.
10. Remove the duplicate rows if any.
11. Identify columns stored with incorrect data types.
12. Convert character variables into factors if any.
13. Convert text values into lowercase/ uppercase.
14. Detect outliers using boxplots.
15. Identify outliers using the interquartile range method.
16. Remove extreme outliers from the dataset.
17. Create a histogram for a numerical variable and interpret its distribution.
18. Add a density curve to the histogram.
19. Create a violin plot
20. Create a dot plot for a small numerical dataset.
21. Create a frequency table for a categorical variable.
22. Calculate the proportion and percentage of each category.
23. Create a bar chart showing the frequency of each category.
24. Create a horizontal bar chart for a variable with several categories.
25. Create a percentage bar chart instead of a frequency bar chart.
26. Create a pie chart for a categorical variable with a small number of categories.
27. Create a lollipop chart for categorical frequencies.
28. Create a dot chart
29. Create a density plot using ggplot2.
30. Use facet_wrap() to display the distribution of one variable across different groups while interpreting each distribution separately.